

CloudUGW

V100R020C60LRUS01SPC100

5G NSA User Traffic Charging

1.1 4G/5G Traffic Differentiation Feature Description

Applicable NEs

S-GW, P-GW

Definition

As defined in 3GPP specifications, the non-standalone (NSA) network leverages existing network resources and enables 4G eNodeBs and 5G NR gNodeBs to share the EPC or 5G core network (5GC), to smoothly evolve from 4G to 5G. In early 5G deployments, the NSA network enables control information to be transmitted from the 5G access network (AN) to the EPC through the 4G AN, so that the 5G air interface can be deployed commercially even if the 5GC specifications have not been finalized. The NSA network makes full use of existing 4G networks and accelerates time to market (TTM) of 5G. In large deployments of 5G, it is foreseeable that NSA helps transmit control information from the 4G AN to the 5GC through the 5G AN, providing carriers with an opportunity to replace 4G networks.

After NSA is deployed on the network, customers want to use the features to differentiate 4G and 5G traffic. Therefore, the following feature needs to be deployed on the MME, S-GW, and P-GW on the NSA network.

The MME dynamically learns eNodeB addresses and automatically maintains the eNodeB address list for identifying NSA areas. The MME notifies the S-GW/PGW that the subscriber is in the 4G or 5G area through a private extension IE. The PGW triggers the RAT change event to the OCS/PCRF to implement differentiated charging for NSA services. The new RAT type can also be recorded in the CDR.

NOTE

This is a customized feature. If it does not meet commercial use requirements, submit a requirement for customization.

Benefits

Beneficiary	Benefits
Carrier	Carriers can accurately identify NSA users and design independent charging packages, which can quickly reclaim carriers' existing network construction investments.
User	This feature is invisible to users.

Availability

Involved NEs

NEs	Supporting Version	Function
S-GW/P-GW	CloudUGW	Identifies 5G NSA UEs based on MME

NEs	Supporting Version	Function
	V100R020C60 SPH125 and later UGW9811 V900R019C10 SPH159 and later	information. A specific NAT type is reported to the PCRF/OCS/CG for NSA UEs.
MME	- CloudUSN V100R020C60 SPH129 and later - USN9810 V900R019C10 SPH317 and later	The gNodeB identifies users and sends the information to the S-GW/P-GW.
PCRF	Versions that support the identification of a new RAT type	Performs charging or bandwidth management based on the RAT type of NSA UEs.
OCS	Versions that support the identification of a new RAT type	The OCS performs online charging based on the RAT type of NSA subscribers.
DRA	Versions that support the identification of a new RAT type	Forwards Diameter messages.
AAA server	Versions that support the identification of a new RAT type	The AAA accounting server performs accounting based on the RAT type of NSA UEs.
CG	- CG9812 V500R006C20 SPC305 and later - CloudCG V100R019C15 SPC500 and later	The CG performs offline charging based on the RAT type of NSA subscribers.

Requirement for the License

This feature is an optional feature and requires a license. The license control items for this feature are "82200DEB LKV3W9UTCH02 5G NSA User Traffic Charging" and "82200DEC LKV3W9UPCO02 5G NSA User Policy Control".

Interaction with Other Features

Interaction Type	Features	Control Item Name	Interaction
Dependency	-	82209469 LKV3W9NBS501 Bearer Numbers of Supporting 5G NSA Basic Software for SGW 82209470 LKV3W9NBP501 Bearer Numbers of Supporting 5G NSA Basic Software for PGW 82209471 LKV3W9NBSP51 Bearer Numbers of Supporting 5G NSA Basic Software for SGW&PGW	The CloudUGW can identify NSA subscribers only when it can access the network. Therefore, you need to apply for the license for the number of NSA subscribers.
Dependency	GWFD-110302 Basic Content Based Charging	N/A.	The flow-based charging function must be enabled on the CloudUGW because the CloudUGW needs to identify services for charging.

Impact on the System

This feature has no impact on the system.

Application Limitations

- This feature involves customized private IEs and can be used only by Huawei MME/S-GW/P-GW. Otherwise, services on the local VNF and peripheral VNFs may be abnormal.
- The CG/AAA server/PCRF/OCS/DRA should identify the extended RAT type (extended RAT type). Confirm that services will not be affected if the RAT type is added for the interworking between the CG/AAA server/PCRF/OCS/DRA and peripheral VNFs.
- This feature does not take effect when the eNodeB and gNodeB on the RAN side use the same user-plane IP address.
- After this feature is enabled, it shall take a period of time to accurately identify UE types. The MME needs to learn eNodeB IP addresses based on the eNodeB user-plane IP address carried in the Initial Context Setup Response message. Currently, the feature takes effect **one day** after the feature is enabled.
- This feature cannot be used for NSA UEs that are performing services when this feature is enabled. The UEs can be identified only after they enter the Active state again.

- If the eNodeB is disconnected for more than three days, the data-plane IP address of the eNodeB stored in the system will be aged. In this case, the system needs to learn the data-plane IP address of the eNodeB contained in the Initial Context Setup Response message in the Attach, TAU, and Service Request procedures again.
- When an NSA-capable UE accesses the network through one of attach, TAU, X2-based handover, and S1-based handover procedures and the UE is within the coverage of a 4G base station, the Create Session Request message sent to the S-GW/P-GW carries the DCNR flag. However, because the proprietary IE can be transmitted to the P-GW through only a Modify Bearer Request message. During this period, 5G PGW-CDRs may be generated.
- During an S1-based handover procedure, the ULI IE in the Create Session Request message carries only the TAI but not the ECGI.
- If handover or location update frequently occurs for 5G NSA UEs, there is a higher probability that signaling conflicts occur on the S-GW/P-GW. To resolve the deadlock caused by the inter-NE signaling conflicts, the Modify Bearer Request message carrying 4G/5G network changes may be optimized or discarded. As a result, 4G/5G network changes of a small number of 5G NSA UEs may be reported to peripheral NEs (PCRF/OCS server/CloudCG/AAA server/DRA) later than expected.
- After this feature is enabled, if the customer is concerned about the consistency between the types of the RAT and ULI in a CDR, you are advised to set **BIT922** to **0** so that the ULI filled in a CDR remains unchanged when the CDR is opened and is the same as the RAT.

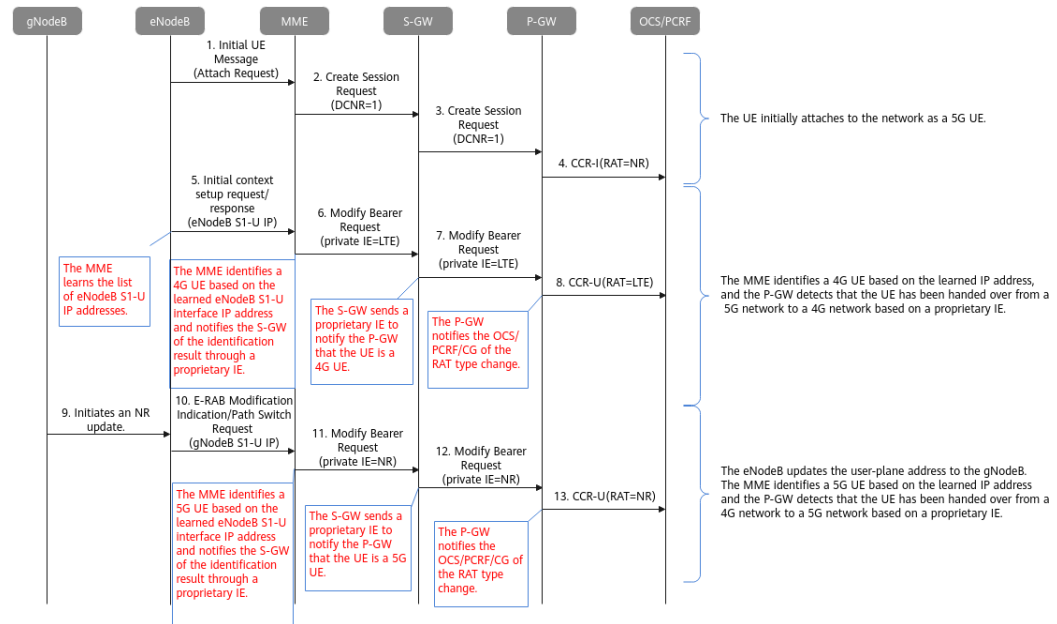
Implementation Principle Feature Description

System Implementation

The MME dynamically learns eNodeB addresses and automatically maintains the eNodeB address list. When a UE moves between a 4G network and a 5G network, the MME checks whether the RAN address is in the eNodeB address list. If the RAN address is not in the eNodeB address list, the MME considers the RAN address as a 5G area. The MME notifies the S-GW/PGW that the subscriber is in the 4G or 5G area. The PGW triggers a RAT change event to the OCS/PCRF to differentiate NSA services.

Service process

Figure 1-1 Flow-based charging for NSA subscribers



Steps 1-4. A UE initially attaches to the network as a 5G UE.

Step 5. After receiving an Initial context setup request message, the eNodeB assigns an S1-U interface address and sends it to the MME through an Initial Context Setup Response message. The MME learns the user-plane IP address of the eNodeB from the response message.

Steps 6-8. The MME identifies a 4G UE based on the learned IP address, and the P-GW detects that the UE has been handed over from a 5G network to a 4G network based on a proprietary IE.

Steps 9-12. The eNodeB updates the user-plane address to the gNodeB. The MME identifies a 5G UE based on the learned IP address and the P-GW detects that the UE has been handed over from a 4G network to a 5G network based on a proprietary IE.

Step 13. If the PGW-C receives a new RAT (NR) and performs offline charging, it populates the customized RAT in CDRs to indicate NSA. If the PGW-C performs online charging, it sends a notify RAT (NR) change event to the OCS server. If the PGW-C interworks with the PCRF, it sends a notify RAT (NR) change event to the PCRF.

Feature Specifications

This feature does not involve feature specifications.

Standard Compliance

This feature does not have any standard restrictions.

Release History

Feature Version	Product Version	Details
3	CloudUGW	Updated the configuration parameters in sections 1.1.1

Feature Version	Product Version	Details
	V100R020C60SPH125	Activating 4G/5G Traffic Differentiation and 1.1.3 Deactivating 4G/5G Traffic Differentiation on the CloudUGW.
2	CloudUGW V100R020C60SPH125	Updated the NE versions and restrictions.
1	CloudUGW V100R020C60	First release.

1.1.1 Activating 4G/5G Traffic Differentiation

Operation Scenario

If carriers plan independent charging for NSA subscribers, activate 4G/5G Traffic Differentiation on the CloudUGW.



NOTE

Applicable to the S-GW and P-GW.

Prerequisites

Prerequisites

- You have familiarized yourself with 1.1 4G/5G Traffic Differentiation Feature Description.
- You have loaded the required licenses.
- The activation is complete.

Data

Category	Parameter	Example Value	How to Obtain	Description
Setting the License Switch	Item	LKV3W9UTCH02 LKV3W9UPCO02	Fixed value	SET LICENSESWITCH
	Switch	ENABLE	Fixed value	
Setting NSA Control Processing	Method for Identifying NSA UEs	AUTOSTUDYNR	Planned on the local end	SET NSACTRLPROP
	NR Flag Switch	ENABLE	Planned on the	

Category	Parameter	Example Value	How to Obtain	Description
			local end	
Configure the RAT value for the subscriber.	Requested RAT type	NR	Fixed value	SET SERVICERATVALUE NOTE Before the configuration, confirm the following items with the peer end: - Whether the peer end can properly process the RAT value NR. - Whether there is special processing for the RAT value NR. The impact brought by the addition of the RAT type must be confirmed.
	RAT value for AAA Accounting	NR	Negotiated with the peer device	
	RAT value used for interworking with the CG	NR	Negotiated with the peer device	
	RAT value used for interworking with the OCS.	NR	Negotiated with the peer device	
	RAT value used for interworking with the PCRF	NR	Negotiated with the peer device	
Configuring Whether to Update the DCNR Flag in Handover Scenarios	Data Type	BIT	Fixed value	SET SOFTPARA
	BIT number	296	Fixed value	
	BIT value	1	Fixed value	
Configuring Whether to Ignore the DCNR Flag for the Specified APN	APN	ims	Planned on the local end	SET APNSOFTPARA
	Data Type	BIT	Fixed value	
	BIT number	35	Fixed value	
	BIT value	1	Planned on the local end	

Procedure

- In the MML Command - UGW_VNFC window of the CGW, run the following command to enable the license:
 SET LICENSESWITCH:ITEM="LKV3W9UTCH02",SWITCH=ENABLE;

- SET LICENSESWITCH:ITEM="LKV3W9UPCO02",SWITCH=ENABLE;
2. In the MML Command - UGW_VNFC window of the CloudUGW, run the following command to enable NSA self-learning and identification:
SET NSACTRLPROP:NSAIDENTIFYMD=AUTOSTUDYNR,
SGWTOPGWNRSW=ENABLE;
 3. In the MML Command - UGW_VNFC window of the CloudUGW, run the following command to specify the RAT type of the NAS for the CG/AAA accounting server/OCS/PCRF:
SET SERVICERATVALUE: REQUESTRAT= *RAT Type*, AAAACCTFORNBLOT=*RAT Type*, CGFORNR= *RAT Type*, OCSFORNR= *RAT Type*, PCRFFORNR= *RAT Type*;
 4. In the MML Command - UGW_VNFC window of the CGW, run the following command to enable DCNR flag update in handover scenarios:
SET SOFTPARA:DT=BIT,BITNUM=296,BITVALUE=1;
 5. (Optional) If differentiated charging is not required for VoLTE services, run the following command in the MML Command - UGW_VNFC window to ignore the DCNR flag for the corresponding APN (APN **ims** is used as an example):

NOTE

If the virtual APN function is enabled, enable the function controlled by the software parameter BIT35 for both the actual APN and virtual APN.

SET APNSOFTPARA:APN="ims",DT=BIT,BITNUM=35,BITVALUE=1;

Verification

For details, see Commissioning 4G/5G Traffic Differentiation on the CloudUGW.

Example

Task Description

Activating 4G/5G Traffic Differentiation on the CloudUGW. Configure the RAT type to be reported to the OCS as NR if the requested RAT type is NR.

NOTE

Before the configuration, confirm the following items with the peer end:

- Whether the peer end can properly process the RAT value NR.
- Whether there is special processing for the RAT value NR. The impact brought by the addition of the RAT type must be confirmed.

Script

1. Enable the license.

```
SET LICENSESWITCH:ITEM="LKV3W9UTCH02",SWITCH=ENABLE;
SET LICENSESWITCH:ITEM="LKV3W9UPCO02",SWITCH=ENABLE;
```

2. Enable the self-learning function to identify NSA UEs.

```
SET NSACTRLPROP:NSAIDENTIFYMD=AUTOSTUDYNR, SGWTOPGWNRSW=ENABLE;
```

3. Configure the RAT type to be reported to the OCS as NR.

```
SET SERVICERATVALUE:REQUESTRAT=NR,OCSFORNR=NR;
```

4. Enable DCNR flag update in handover scenarios.

```
SET SOFTPARA:DT=BIT,BITNUM=296,BITVALUE=1;
```

- (Optional) If differentiated charging is not required for VoLTE services, ignore the DCNR flag for the corresponding APN (APN **ims** is used as an example).

NOTE

If the virtual APN function is enabled, enable the function controlled by the software parameter BIT35 for both the actual APN and virtual APN.

```
SET APNSOFTPARA:APN="ims",DT=BIT,BITNUM=35,BITVALUE=1;
```

Example

Task Description

Activating 4G/5G Traffic Differentiation on the CloudUGW. Configure the RAT type to be reported to the CG/AAA accounting server/OCS/PCRF as NR if the requested RAT type is NR.

NOTE

Before the configuration, confirm the following items with the peer end:

- Whether the peer end can properly process the RAT value NR.
- Whether there is special processing for the RAT value NR. The impact brought by the addition of the RAT type must be confirmed.

Script

- Enable the self-learning function to identify NSA UEs.

```
SET NSACTRLPROP:NSAIDENTIFYMD=AUTOSTUDYNR, SGWTOPGWNRSW=ENABLE;
```

- Enable the self-learning function to identify NSA UEs.

```
SET NSACTRLPROP:NSAIDENTIFYMD=AUTOSTUDYNR, SGWTOPGWNRSW=ENABLE;
```

- Configure the RAT type to be reported to the CG/AAA accounting server/PCRF/OCS as NR.

```
SET  
SERVICERATVALUE:REQUESTRAT=NR, AAAACCTFORNR=NR, CGFORNR=NR, OCSFORNR=NR, PCRFFORNR=  
NR;
```

- Enable DCNR flag update in handover scenarios.

```
SET SOFTPARA:DT=BIT,BITNUM=296,BITVALUE=1;
```

- (Optional) If differentiated charging is not required for VoLTE services, ignore the DCNR flag for the corresponding APN (APN **ims** is used as an example).

NOTE

If the virtual APN function is enabled, enable the function controlled by the software parameter BIT35 for both the actual APN and virtual APN.

```
SET APNSOFTPARA:APN="ims",DT=BIT,BITNUM=35,BITVALUE=1;
```

1.1.2 Commissioning 4G/5G Traffic Differentiation on the CloudUGW

Scenario

Commission this feature when the NSA feature is deployed and independent charging is required for NSA subscribers to ensure that independent charging takes effect.

NOTE

Applicable to the S-GW and P-GW.

Prerequisites

Prerequisites

- You have familiarized yourself with 1.1 4G/5G Traffic Differentiation Feature Description.
- 1.1.1 Activating 4G/5G Traffic Differentiation is complete.

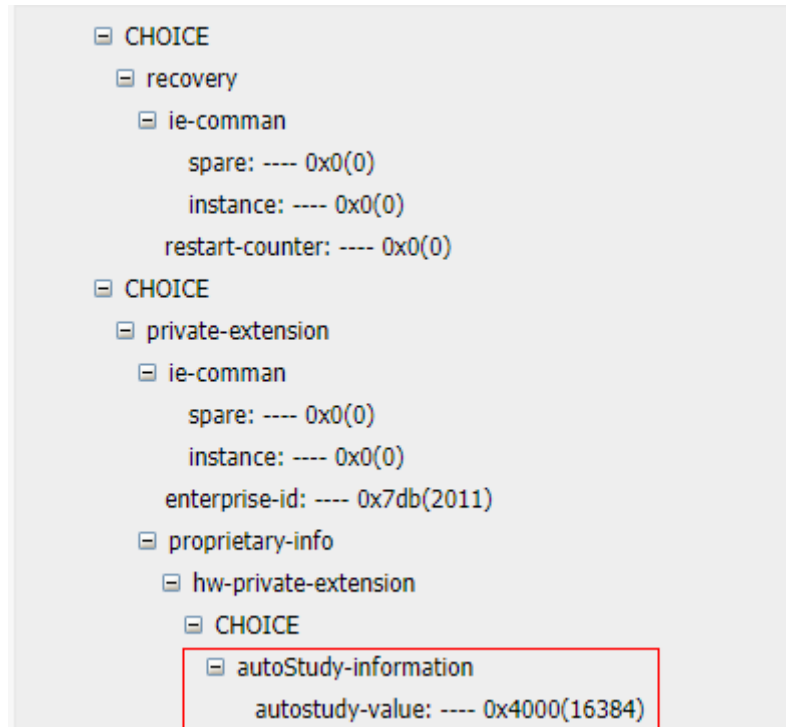
Data

Category	Parameter	Example Value	How to Obtain	Description
User information query	IMSI	460000123456789	Built-in with the test terminal	-
APN used by the test terminal	APN (APN)	test	Obtained from the configured data	This parameter is set to the APN instance name configured in Configuring a Common APN.

Procedure

1. Create a GGSN+P-GW+S-GW user tracing task on the WebLMT and select the default message type for Message Type.
2. Use the test UE to access the network using the APN test.
 - If the test MS/UE successfully accesses the network, go to Step 4.
 - If the test terminal cannot access the network, commission the S-GW/P-GW access function.
3. Use the test MS/UE to browse web pages.
4. View the traced messages and check the reported messages. Whether the message contains a private extension message as following shown in Figure 1-2.

Figure 1-2 User tracing



- If yes, no further action is required.
- If no, go to 5.
- 5. Collect information and seek technical support.
 - a. Create a user tracing task on the WebLMT and run the following step 4 and save the packet.
 - b. Run the command to export the current configuration data as an MML script file and save the file.
 - c. Collect and save the preceding information.
 - d. Collect and summarize all information and contact Huawei technical support engineers.

1.1.3 Deactivating 4G/5G Traffic Differentiation on the CloudUGW

Scenario

If carriers plan not to charge NSA UEs separately, disable 4G/5G Traffic Differentiation feature.

NOTE

Applicable to the S-GW and P-GW.

Prerequisites

Prerequisites

- 1.1.1 Activating 4G/5G Traffic Differentiation is complete.
- The configuration file has been backed up locally.

Data

Category	Parameter	Example Value	How to Obtain	Description
Setting NSA Control Processing	Method for Identifying NSA UEs	DCNR	Planned on the local end	SET NSACTRLPROP
	NR Flag Switch	DISABLE	Planned on the local end	
Configure the RAT value for the subscriber.	Requested RAT type	NR	Fixed value	SET SERVICERATVALUE
	RAT value for AAA Accounting	EUTRAN	Planned on the local end	
	RAT value used for interworking with the CG	EUTRAN	Planned on the local end	
	RAT value used for interworking with the OCS.	EUTRAN	Planned on the local end	
	RAT value used for interworking with the PCRF	EUTRAN	Planned on the local end	
Configuring Whether to Update the DCNR Flag in Handover Scenarios	Data Type	BIT	Fixed value	SET SOFTPARA
	BIT number	296	Fixed value	
	BIT value	0	Fixed value	

Procedure

1. In the MML Command - UGW_VNFC window of the CloudUGW, disable NSA self-learning and identification.

```
SET NSACTRLPROP:NSAIDENTIFYMD=DCNR, SGWTOPGWNRSW=DISABLE;
```

2. In the MML Command - UGW_VNFC window of the CloudUGW, configure the CG/AAA accounting server/OCS/PCRF to carry the NAS RAT type.

```
SET SERVICERATVALUE: REQUESTRAT= RAT Type, AAAACCTFORNR= RAT Type, CGFORNR= RAT Type, OCSFORNR= RAT Type, PCRFFORNR= RAT Type;
```

3. In the MML Command - UGW_VNFC window of the CGW, run the following command to disable DCNR flag update in handover scenarios:

```
SET SOFTPARA:DT=BIT,BITNUM=296,BITVALUE=0;
```

Verification

1. In the MML Command - UGW window, run LST NSACTRLPROP. Check whether the values of **The way to identify NSA users** and **NR Flag Switch** are **DCNR** and **DISABLE**, respectively.

```
LST NSACTRLPROP;;
NSA Control Process Configuration
-----
                The way to identify NSA users = DCNR
                        NR Flag Switch = DISABLE
                CloudUGW Supports 5G Traffic Reporting = ENABLE
                SGW Accumulates 5G Traffic in S1 Release = DISABLE
```

2. Check whether the RAT value used by neighboring NEs to send messages of the NSA UE is EUTRAN.

3. In the MML Command - UGW window, run LST SERVICERATVALUE to list the RAT value used when the CloudUGW sends messages to a peer end.

```
LST SERVICERATVALUE;; %%
RETCODE = 0 Operation Success.

User RAT Setting
-----
Request Rat Type RAT value for AAA Authentication RAT value for AAA Accounting RAT
value for CG RAT value for OCS RAT value for PCRF

EUTRAN-NB-IoT EUTRAN-NB-IoT EUTRAN-NB-IoT EUTRAN-NB-IoT EUTRAN-NB-IoT
EUTRAN-NB-IoT
LTE-M EUTRAN EUTRAN EUTRAN EUTRAN EUTRAN
NR EUTRAN EUTRAN EUTRAN EUTRAN EUTRAN
(Number of results = 3)
--- END
```

Example

Task Description

Deactivating 4G/5G Traffic Differentiation on the CloudUGW.

Script

1. The eNodeB does not identify NSA UEs using self-learning.

```
SET NSACTRLPROP:NSAIDENTIFYMD=DCNR, SGWTOPGWNRSW=DISABLE;
```

2. Configure the RAT type to be reported to the CG/AAA accounting server/PCRF/OCS as E-UTRAN.

```
SET  
SERVICERATVALUE:REQUESTRAT=NR,AAAACCTFORNR=EUTRAN,CGFORNR=EUTRAN,OCSFORNR=EUTRAN,PCRFFORNR=EUTRAN;
```

3. Disable DCNR flag update in handover scenarios.

```
SET SOFTPARA:DT=BIT,BITNUM=296,BITVALUE=0;
```